Chapter 3: Logistic, Ridge, and Lasso Regression

Part II — Introduction to Statistical Learning

Omid Safarzadeh

Abstract

This chapter extends linear regression in two directions. *Logistic regression* swaps the continuous output for a probability, turning the linear model into a classifier whose loss is cross-entropy. *Ridge* and *lasso* add a penalty on the coefficients to trade a little bias for a large cut in variance — ridge by shrinking every coefficient, lasso by setting some to exactly zero. We derive each, prove that the logistic loss is convex (so there is a single optimum), write ridge in closed form, and show why the lasso's soft-threshold is what produces sparsity.

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1 From a line to a probability, and a penalty on the fit

To classify rather than predict a number, we model the *probability* of the positive class as a linear function squashed into $(0, 1)$ by the **sigmoid** $\sigma(z) = 1/(1 + e^{-z})$:

$$P(y = 1 \mid x) = \sigma(\beta^\top x), \qquad \log \frac{P(y = 1 \mid x)}{P(y = 0 \mid x)} = \beta^\top x.$$

The log-odds are linear in x, so the decision boundary $\beta^\top x = 0$ is a hyperplane. Fitting is maximum likelihood, and the negative log-likelihood is the **cross-entropy** loss

$$\mathcal{L}(\beta) = -\sum_{i=1}^{n} \Big[y_i \log \sigma(\beta^\top x_i) + (1 - y_i) \log \big(1 - \sigma(\beta^\top x_i)\big) \Big].$$

Separately, when predictors are many or collinear, ordinary least squares overfits. **Regularization** adds a penalty on the size of β:

$$\text{ridge: } \min_\beta \|y - X\beta\|^2 + \lambda \|\beta\|_2^2, \qquad \text{lasso: } \min_\beta \|y - X\beta\|^2 + \lambda \|\beta\|_1.$$

The squared (ℓ_2) penalty shrinks coefficients smoothly; the absolute (ℓ_1) penalty shrinks them *and* drives small ones to exactly zero, so the lasso also selects variables. The building blocks are short:

```
import math

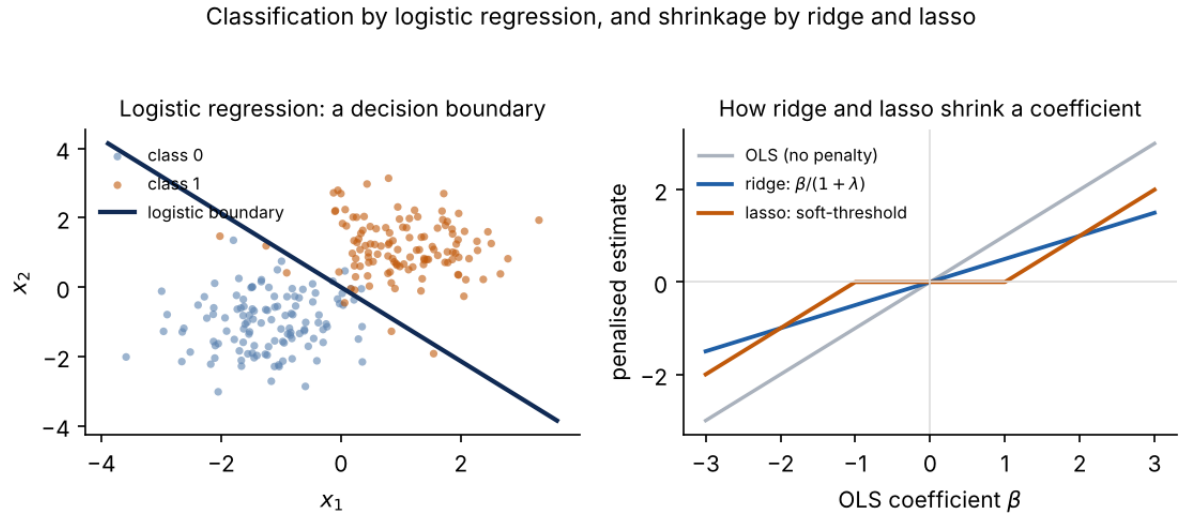
def sigmoid(z):
    return 1.0 / (1.0 + math.exp(-z))
```

```python
def cross_entropy(y, p):                          # one observation
    return -(y * math.log(p) + (1 - y) * math.log(1 - p))

def soft_threshold(z, lam):                       # the lasso's shrinkage
    if z >  lam: return z - lam
    if z < -lam: return z + lam
    return 0.0

print(round(sigmoid(0.0), 3))                     # 0.5
print(round(cross_entropy(1, 0.9), 4))            # 0.1054
print(soft_threshold(0.7, 1.0), soft_threshold(2.0, 1.0))   # 0.0 1.0
```

The figure shows both ideas: a logistic decision boundary separating two classes, and the three shrinkage rules side by side — OLS leaves a coefficient alone, ridge scales it down, and the lasso's soft-threshold flattens a whole band around zero flat to zero.



2 Why these fits are well-behaved

Theorem 2.1 (The logistic loss is convex)**.** The cross-entropy $\mathcal{L}(\beta)$ is convex in β; its Hessian is positive semidefinite.

Proof. Let $p_i = \sigma(\beta^\top x_i)$. The gradient is $\nabla\mathcal{L} = \sum_i (p_i - y_i)\, x_i = X^\top(p - y)$, and since $\sigma'(z) = \sigma(z)(1-\sigma(z))$, the Hessian is

$$\nabla^2\mathcal{L} = \sum_i p_i(1-p_i)\, x_i x_i^\top = X^\top S X, \qquad S = \operatorname{diag}\big(p_i(1-p_i)\big).$$

Each $p_i(1-p_i) \geq 0$, so $S \succeq 0$ and $X^\top S X \succeq 0$. A twice-differentiable function with a positive-semidefinite Hessian is convex, so the loss has no local minima to get stuck in — gradient descent finds the global optimum. □

Ridge, unlike the lasso, has a closed form, and it fixes a failure of OLS.

Theorem 2.2 (Ridge solution)**.** $\hat\beta_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$, and the matrix $X^\top X + \lambda I$ is invertible for every $\lambda > 0$ even when $X^\top X$ is singular.

Proof. Differentiate $\|y - X\beta\|^2 + \lambda\|\beta\|^2$ and set to zero: $-2X^\top(y - X\beta) + 2\lambda\beta = 0$, giving $(X^\top X + \lambda I)\beta = X^\top y$. For any $\lambda > 0$ the matrix $X^\top X + \lambda I$ has all eigenvalues $\geq \lambda > 0$

(those of $X^\top X$ are ≥ 0), so it is invertible — ridge is defined even with more predictors than observations. □

The lasso has no such formula in general, but for one coordinate at a time it does, and that formula is exactly the soft-threshold.

Theorem 2.3 (Lasso's soft-threshold)**.** Minimising $\frac{1}{2}(z-\beta)^2+\lambda|\beta|$ over β gives $\hat{\beta} = S_\lambda(z) = \operatorname{sign}(z)\,\max(|z| - \lambda,\, 0)$.

Proof. For $\beta > 0$ the objective is smooth with derivative $\beta - z + \lambda$, zero at $\beta = z - \lambda$ (valid when $z > \lambda$); for $\beta < 0$ similarly $\beta = z + \lambda$ (valid when $z < -\lambda$). When $|z| \leq \lambda$ the subgradient of $|\beta|$ at 0 is the interval $[-1, 1]$ and $0 \in [z-\lambda,\, z+\lambda]$, so $\hat{\beta} = 0$. Collecting the cases gives $S_\lambda(z)$ — which is zero on a whole band $|z| \leq \lambda$, and that flat band is where sparsity comes from. □

3 The geometry of sparsity, and beyond

Why does ℓ_1 zero coefficients out while ℓ_2 does not? The penalty $\|\beta\|_1 \leq t$ is a diamond with corners on the axes, while $\|\beta\|_2 \leq t$ is a round ball. The least-squares contours first touch the diamond at a corner — where some coordinates are exactly zero — whereas they touch the ball on a smooth face, where no coordinate is exactly zero. Sparsity is a consequence of the corners.

The absolute value is not differentiable at zero, so there is no neat normal equation; the optimum is characterised by **subgradients** (the $[-1, 1]$ interval used above) and computed by **coordinate descent**, which applies the soft-threshold to one coefficient at a time until nothing moves. The **elastic net** combines both penalties, $\lambda\big(\alpha\|\beta\|_1 + (1-\alpha)\|\beta\|_2^2\big)$, keeping the lasso's selection while handling groups of correlated predictors more gracefully. All of this buys variance reduction at the price of a little bias — the same trade-off of Chapter 2, now controlled by λ.

Summary

Logistic regression models $P(y = 1 \mid x) = \sigma(\beta^\top x)$ and is fit by minimising the convex cross-entropy loss (Hessian $X^\top S X \succeq 0$). Ridge adds $\lambda\|\beta\|_2^2$, with closed form $(X^\top X + \lambda I)^{-1} X^\top y$ that is always invertible; the lasso adds $\lambda\|\beta\|_1$, whose per-coordinate solution is the soft-threshold $S_\lambda(z) = \operatorname{sign}(z)\max(|z|-\lambda, 0)$, zeroing small coefficients and so selecting variables. The ℓ_1 corners are the geometric reason for sparsity.